

## UECKERMUENDE, Germany

WMO: 101930

Lat: 53.7444N

Lon: 14.0697E

Elev: 7

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 6.1        | 12.9 | -1.2                          | 5.2 | 7.1  | 6.1 | 7.5 | 13.9 | 27.8                  | 46.9 | 25.3 | 43.6 | 3.6                   | 80   | 0.584 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.3                   | 81.3            | 66.1 | 77.5 | 65.0 | 74.1 | 63.3 | 68.4                | 77.1 | 66.6 | 74.9 | 64.7 | 72.4 | 8.8                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.2                            | 93.3 | 72.6 | 63.3 | 87.1 | 71.0 | 61.3 | 80.9 | 68.7 | 32.6          | 76.8 | 31.1 | 75.0 | 29.7 | 72.7 | 75.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-----|--------------------|------|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |      | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max  | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 23.5          | 20.8 | 18.5 | DB                         | 6.4 | 89.7               | 11.9 | 3.6                                                | -2.1 | 92.3       | -9.1 | 94.4       | -15.7 | 96.5       | -24.3 | 99.1 | (4) |
| (5)               |      |      | WB                         | 5.3 | 72.1               | 11.6 | 2.2                                                | -3.0 | 73.7       | -9.8 | 75.0       | -16.3 | 76.2       | -24.8 | 77.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 47.5        | 32.4  | 32.0  | 39.0  | 45.3  | 54.0  | 59.6  | 63.5  | 62.8  | 56.4  | 49.4  | 39.8  | 35.0  | (6)  |  |
|                                                                        | DBStd       | 13.16       | 10.71 | 9.37  | 7.37  | 6.38  | 6.48  | 5.53  | 4.99  | 5.13  | 4.60  | 6.34  | 5.92  | 6.80  | (7)  |  |
|                                                                        | HDD50       | 2461        | 545   | 505   | 345   | 173   | 30    | 1     | 0     | 0     | 3     | 86    | 308   | 465   | (8)  |  |
|                                                                        | HDD65       | 6491        | 1010  | 924   | 807   | 593   | 347   | 181   | 89    | 107   | 262   | 484   | 757   | 930   | (9)  |  |
|                                                                        | CDD50       | 1553        | 0     | 1     | 2     | 30    | 154   | 288   | 418   | 397   | 195   | 67    | 1     | 0     | (10) |  |
|                                                                        | CDD65       | 110         | 0     | 0     | 0     | 1     | 6     | 18    | 42    | 39    | 4     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 754         | 0     | 0     | 0     | 4     | 57    | 135   | 286   | 240   | 31    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 164         | 0     | 0     | 0     | 0     | 7     | 29    | 65    | 58    | 5     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 8.6         | 10.8  | 9.3   | 9.4   | 8.5   | 7.5   | 7.5   | 7.3   | 6.8   | 8.0   | 8.5   | 9.5   | 10.1  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 22.2        | 1.6   | 1.3   | 1.5   | 1.2   | 2.0   | 2.4   | 2.8   | 2.5   | 2.0   | 1.7   | 1.7   | 1.7   | (15) |  |
|                                                                        | PrecMax     | 28.4        | 3.0   | 2.8   | 3.1   | 2.8   | 3.7   | 4.2   | 9.6   | 6.0   | 4.2   | 3.6   | 3.6   | 3.2   | (16) |  |
|                                                                        | PrecMin     | 17.0        | 0.0   | 0.2   | 0.4   | 0.2   | 0.2   | 1.0   | 0.4   | 0.9   | 0.6   | 0.7   | 0.0   | 0.3   | (17) |  |
|                                                                        | PrecStd     | 3.2         | 0.8   | 0.6   | 0.7   | 0.6   | 0.9   | 1.1   | 1.8   | 1.0   | 0.9   | 0.7   | 0.7   | 0.7   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 52.6  | 57.9  | 61.5  | 72.4  | 80.1  | 84.3  | 86.1  | 85.9  | 78.0  | 69.1  | 55.6  | 53.1  | (19) |  |
|                                                                        |             | MCWB        | 48.9  | 50.9  | 51.8  | 56.7  | 63.3  | 65.7  | 67.2  | 67.2  | 64.6  | 61.2  | 51.5  | 48.6  | (20) |  |
|                                                                        | 2%          | DB          | 49.0  | 51.3  | 54.9  | 63.7  | 74.0  | 78.1  | 81.0  | 80.3  | 72.2  | 64.7  | 52.3  | 50.7  | (21) |  |
|                                                                        |             | MCWB        | 45.7  | 46.7  | 47.9  | 52.7  | 59.9  | 65.2  | 66.6  | 66.1  | 62.2  | 57.6  | 49.4  | 47.5  | (22) |  |
|                                                                        | 5%          | DB          | 46.8  | 46.3  | 51.3  | 59.3  | 69.1  | 73.7  | 77.0  | 75.6  | 67.7  | 61.2  | 50.2  | 48.3  | (23) |  |
|                                                                        |             | MCWB        | 43.8  | 42.6  | 45.8  | 50.9  | 59.3  | 63.0  | 65.4  | 64.3  | 60.1  | 55.2  | 47.2  | 45.6  | (24) |  |
|                                                                        | 10%         | DB          | 44.4  | 43.1  | 48.5  | 55.3  | 64.9  | 69.8  | 73.6  | 72.3  | 64.6  | 58.3  | 48.0  | 45.0  | (25) |  |
|                                                                        |             | MCWB        | 41.7  | 39.8  | 43.9  | 48.7  | 56.8  | 60.9  | 63.7  | 63.3  | 58.2  | 53.3  | 45.4  | 42.8  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 48.9  | 51.5  | 52.3  | 57.6  | 64.5  | 69.7  | 71.5  | 70.1  | 66.6  | 62.0  | 52.6  | 49.7  | (27) |  |
|                                                                        |             | MCDB        | 51.8  | 56.1  | 59.9  | 68.3  | 76.2  | 77.7  | 80.2  | 80.5  | 75.5  | 67.8  | 54.3  | 52.3  | (28) |  |
|                                                                        | 2%          | WB          | 46.2  | 46.3  | 49.1  | 54.5  | 62.0  | 66.3  | 68.8  | 67.8  | 63.6  | 58.5  | 49.7  | 47.7  | (29) |  |
|                                                                        |             | MCDB        | 48.5  | 50.8  | 53.5  | 61.7  | 71.8  | 75.5  | 77.3  | 76.5  | 70.3  | 63.9  | 52.1  | 50.4  | (30) |  |
|                                                                        | 5%          | WB          | 44.0  | 42.9  | 46.4  | 51.7  | 59.7  | 63.8  | 66.7  | 66.2  | 61.3  | 55.9  | 47.7  | 45.7  | (31) |  |
|                                                                        |             | MCDB        | 46.4  | 45.8  | 51.1  | 58.5  | 67.8  | 71.8  | 74.8  | 74.1  | 66.3  | 60.3  | 49.9  | 48.0  | (32) |  |
|                                                                        | 10%         | WB          | 41.7  | 40.2  | 44.0  | 49.6  | 57.5  | 61.7  | 64.7  | 64.2  | 59.1  | 53.8  | 45.7  | 42.8  | (33) |  |
|                                                                        |             | MCDB        | 44.2  | 42.9  | 48.0  | 55.1  | 64.1  | 68.9  | 72.1  | 71.3  | 64.0  | 57.5  | 47.7  | 44.8  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.6   | 9.8   | 11.3  | 13.7  | 14.8  | 14.1  | 15.3  | 15.2  | 13.8  | 12.2  | 8.4   | 7.4   | (35) |  |
|                                                                        |             | MCDBR       | 8.7   | 12.3  | 16.0  | 20.0  | 21.3  | 21.3  | 21.6  | 21.2  | 19.4  | 15.8  | 9.8   | 9.5   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 7.8   | 9.3   | 10.9  | 11.9  | 12.3  | 12.2  | 11.7  | 11.8  | 11.8  | 10.6  | 8.9   | 8.4   | (37) |  |
|                                                                        |             | MCWBR       | 8.7   | 12.5  | 15.6  | 18.6  | 19.3  | 18.4  | 19.2  | 18.3  | 17.3  | 13.9  | 9.4   | 8.8   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.317 | 0.342 | 0.373 | 0.409 | 0.407 | 0.403 | 0.424 | 0.420 | 0.385 | 0.364 | 0.342 | 0.312 | (40) |  |
|                                                                        |             | taud        | 2.325 | 2.317 | 2.264 | 2.228 | 2.285 | 2.324 | 2.280 | 2.299 | 2.374 | 2.391 | 2.367 | 2.320 | (41) |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 203   | 237   | 253   | 259   | 267   | 269   | 260   | 253   | 248   | 226   | 192   | 180   | (42) |  |
|                                                                        |             | Edh at Noon | 19    | 26    | 33    | 39    | 39    | 38    | 39    | 36    | 30    | 24    | 18    | 16    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 176   | 384   | 769   | 1313  | 1650  | 1744  | 1616  | 1357  | 987   | 516   | 215   | 122   | (44) |  |
|                                                                        | RadStd      | RadStd      | 19    | 56    | 85    | 175   | 193   | 135   | 209   | 113   | 104   | 73    | 28    | 11    | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) Regional (42 neighbors) |  | +0.68 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | -263        | N/A   | N/A   |       |

Nomenclature: See separate page